

Arboretum Eco-Quest

Solve outdoor ecology checkpoints

At a glance: Read the living collection and solve the checkpoints before time runs out. Plan about 80 minutes for the core block; 4 teams of 4 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: tree identification, ecosystems, observation, classification evidence (Understand).
- Plan an efficient checkpoint route that stays within approved field boundaries (Apply).
- Use observable traits to narrow possibilities, then use a staff-verified field tag, visible label, or route key to confirm species identity (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- 4 sets of clue cards, one card per checkpoint (plus 1 spare set; card set in this guide)
- 1 laminated adaptive evidence key per team (plus 1 spare); use a staff-verified numbered field tag, visible label, or route key to confirm any species name
- 4 route maps with the checkpoints marked (plus 2 spares; route map in this guide)
- 4 clipboards (plus 1 spare)
- 4 evidence token bags (plus 1 spare), with evidence tokens for each checkpoint (about 30 tokens; cut-apart token sheet in this guide)
- pencils, shared
- optional QR codes that open the Arboretum Explorer record at each checkpoint

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

- 01 Assemble one route pack per team with the materials above, sorted and ready.
- 02 Stage the scoring sheet, staff route key, and verified identity references before teams begin.
- 03 No PPE: this is an outdoor observation walk with no chemicals, water, soil, or instruments. Set up sun and hydration support, run the weather and heat-index check, confirm the buddy system and route boundaries, and confirm the approved route with the arboretum host.
- 04 Load the activity slides and ready the projected timer.

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Read the living collection and solve the checkpoints before time runs out." Take a quick prediction by show of hands.

Phase 2 Predict and plan the route

0:05 - 0:12

Each team takes a quick prediction about which clue will be most reliable, then numbers the checkpoints on the route map in an order that avoids backtracking. No walking yet.

Phase 3 Constraints

0:12 - 0:18

Set the field method: stay with the team on approved paths, plan a no-backtrack route, record observable traits at every checkpoint, and use verified sources for species names.

Phase 4 Walk and solve

0:18 - 0:44

Teams walk their route, use the clue cards and adaptive evidence key to record observable traits, then use a staff-verified numbered field tag, visible label, or route key for any species name. They record the evidence and collect the token. Circulate and ask which evidence supports each answer.

Phase 5 Record evidence and collect tokens

At each checkpoint, teams write the one specific clue that justifies their answer and take the evidence token.

Phase 6 Check answers

Before returning, each team confirms every answer cites a real observed clue, not a guess, and has a token for each solved checkpoint.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, and reset the station. Collect the route packs, the keys, the clipboards, and the evidence tokens; there is no PPE to remove and nothing to wash.

TROUBLESHOOTING

Problem: A team backtracks and wastes time. **Fix:** Have them re-number the remaining checkpoints into a no-backtrack order; treat it as a route-planning coaching moment, since pace is scored.

Problem: A team is stuck at a checkpoint. **Fix:** Have them use the adaptive evidence key to record the observable leaf or bark trait. Confirm any species name only from a staff-verified numbered field tag, visible label, route key, or verified Arboretum Explorer record.

Problem: A team answers from memory instead of the clues. **Fix:** Require them to write the specific feature (leaf shape, bark, branching, or seed) for each answer; an answer with no cited clue scores zero on evidence quality.

Problem: A team finishes early. **Fix:** Ask them to recheck one checkpoint with a second observable trait and strengthen their evidence record.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to observation as evidence.

Tier 2 (observing but not explaining):

- "Which observable trait supports your answer, and what source verifies any species name?"

Tier 3 (ready to extend):

- "Recheck one checkpoint with a second observable trait, then verify any species name against a field tag, visible label, staff route key, or Arboretum Explorer record."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Which clue was most reliable for telling trees apart?
- Where did a key send you wrong, and why?
- What makes a good piece of field evidence?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Checkpoint accuracy	50
Evidence quality	20
Pace	20
Teamwork and conduct	10
Total	100

CLEANUP AND SOURCES

Collect the route maps, clue cards, adaptive evidence keys, clipboards, and evidence tokens and return them to the bin, and collect score slips. Nothing is consumed and there are no surfaces to wipe.

SOURCES Ambler Arboretum self-guided tour (<https://ambler.temple.edu/arboretum/our-gardens/self-guided-tour>); Arboretum Explorer interactive collection map (<https://ambler.arboretumexplorer.org/>)